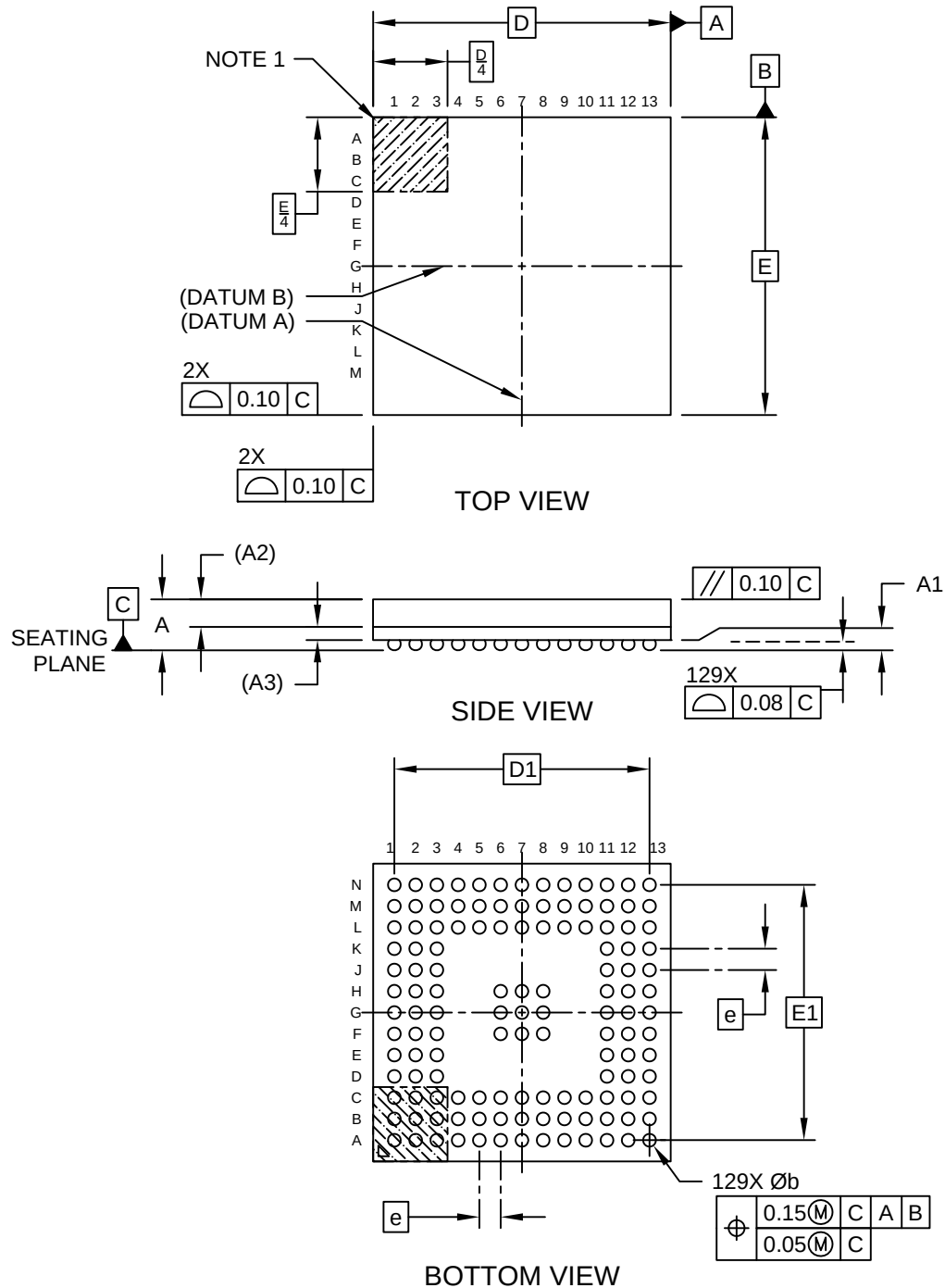


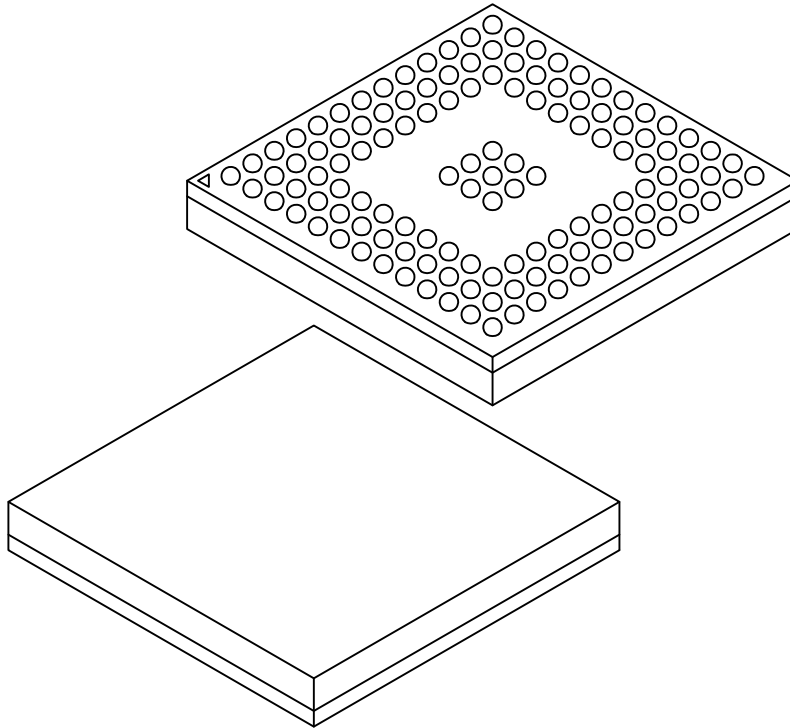
# 129-Ball Thin Fine Pitch Ball Grid Array (GW) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 129-Ball Thin Fine Pitch Ball Grid Array (GW) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



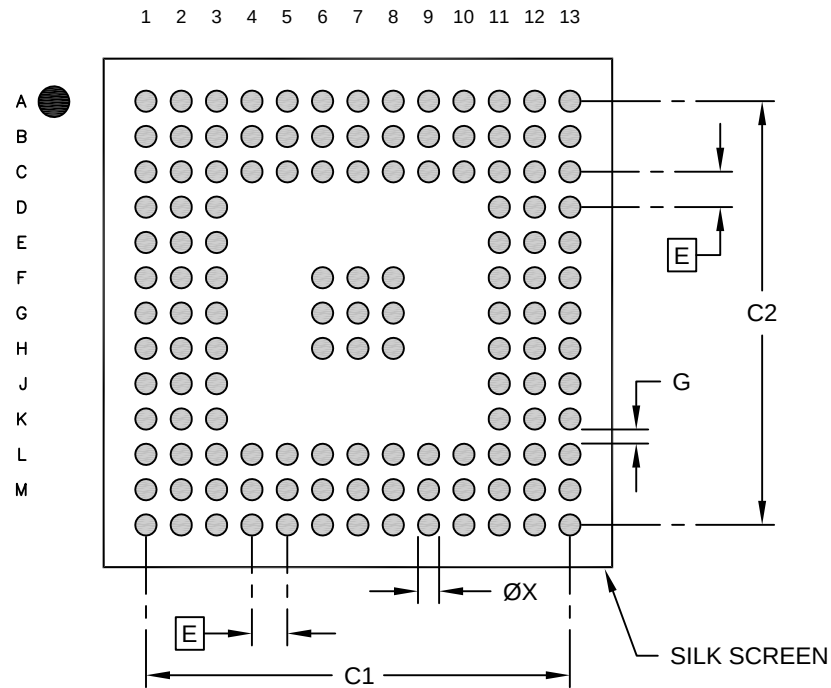
|                          |    | Units | MILLIMETERS |      |      |
|--------------------------|----|-------|-------------|------|------|
| Dimension Limits         |    |       | MIN         | NOM  | MAX  |
| Number of Terminals      | N  |       | 129         |      |      |
| Pitch                    | e  |       | 0.50 BSC    |      |      |
| Overall Height           | A  |       | -           | -    | 1.20 |
| Standoff                 | A1 |       | 0.20        | 0.24 | 0.28 |
| Mold Thickness           | A2 |       | 0.65 REF    |      |      |
| Substrate Thickness      | A3 |       | 0.21 REF    |      |      |
| Overall Length           | D  |       | 7.00 BSC    |      |      |
| Overall Terminal Spacing | D1 |       | 6.00 BSC    |      |      |
| Overall Width            | E  |       | 7.00 BSC    |      |      |
| Overall Terminal Spacing | E1 |       | 6.00 BSC    |      |      |
| Terminal Diameter        | b  |       | 0.25        | 0.30 | 0.35 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 129-Ball Thin Fine Pitch Ball Grid Array (GW) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.50 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 6.00 |      |
| Overall Contact Pad Spacing | C2 |             | 6.00 |      |
| Contact Pad Width (X129)    | X1 |             |      | 0.30 |
| Contact Pad to Contact Pad  | G  | 0.20        |      |      |

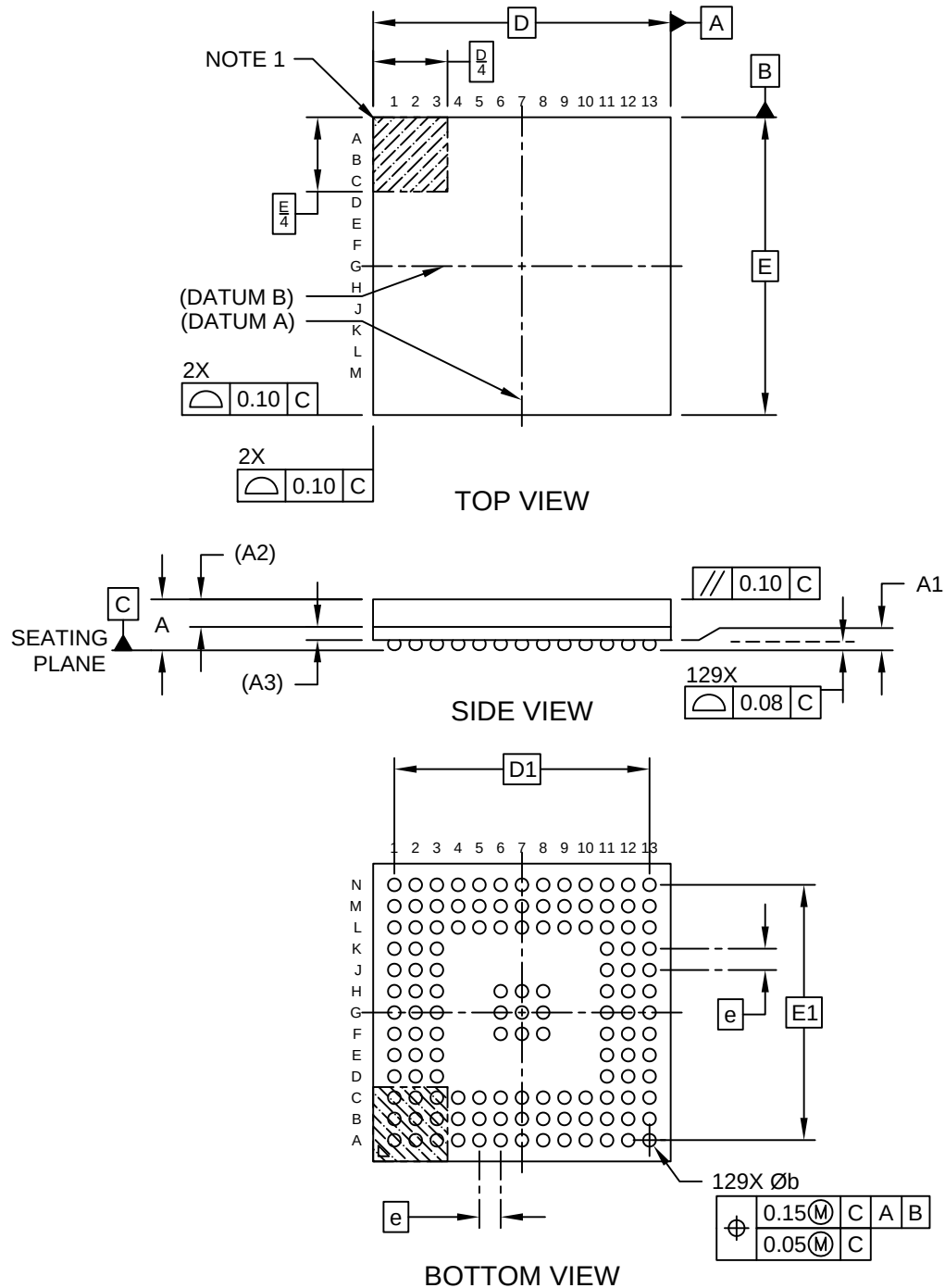
**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

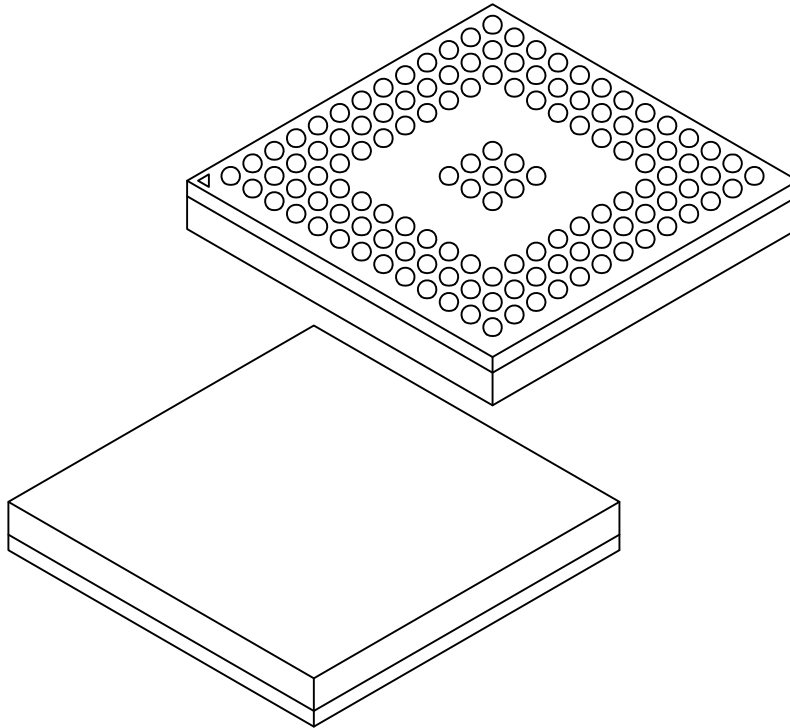
# 129-Ball Thin Fine Pitch Ball Grid Array (2H) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 129-Ball Thin Fine Pitch Ball Grid Array (2H) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



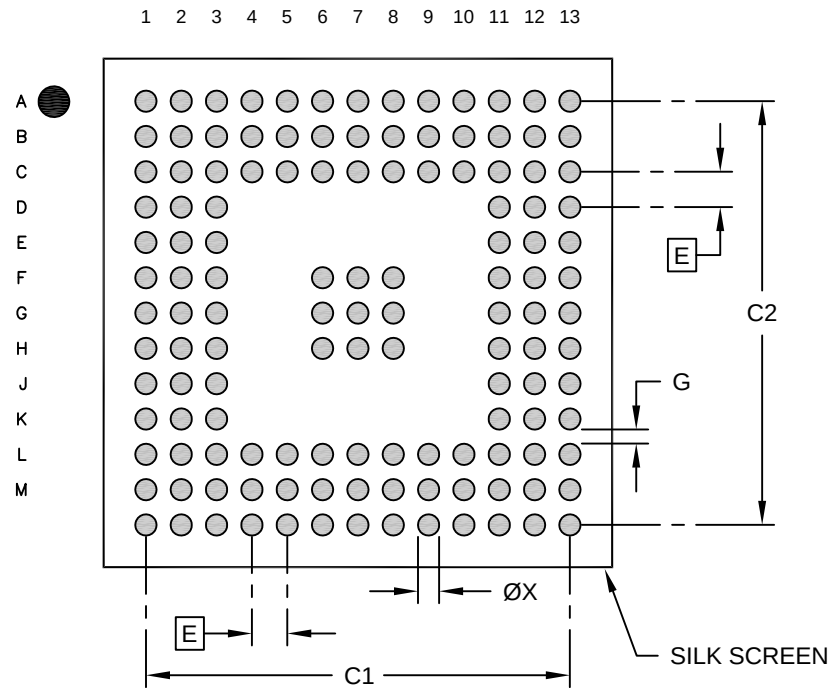
| Units                    |    | MILLIMETERS |      |      |
|--------------------------|----|-------------|------|------|
| Dimension Limits         |    | MIN         | NOM  | MAX  |
| Number of Terminals      | N  | 129         |      |      |
| Pitch                    | e  | 0.50 BSC    |      |      |
| Overall Height           | A  | -           | -    | 1.20 |
| Standoff                 | A1 | 0.20        | 0.24 | 0.28 |
| Mold Thickness           | A2 | 0.65 REF    |      |      |
| Substrate Thickness      | A3 | 0.21 REF    |      |      |
| Overall Length           | D  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | D1 | 6.00 BSC    |      |      |
| Overall Width            | E  | 7.00 BSC    |      |      |
| Overall Terminal Spacing | E1 | 6.00 BSC    |      |      |
| Terminal Diameter        | b  | 0.25        | 0.30 | 0.35 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 129-Ball Thin Fine Pitch Ball Grid Array (2H) - 7x7 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.50 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 6.00 |      |
| Overall Contact Pad Spacing | C2 |             | 6.00 |      |
| Contact Pad Width (X129)    | X1 |             |      | 0.30 |
| Contact Pad to Contact Pad  | G  | 0.20        |      |      |

**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.